Algebra Monsoon 2026 Exercise : the product magma

Let E_1, E_2 be sets.
You know the universal property of the product $(E_1 \times E_2, \pi_1, \pi_2)$.

Using only this universal property (without using the explicit description of $E_1 \times E_2$ as the set of pairs (x_1, x_2), where $x_1 \in E_1$ and $x_2 \in E_2$)
show that, if E_1, E_2 are magmas,

(1) there is a unique law of composition on $E_1 \times E_2$ making π_1, π_2 morphisms of magmas,

(2) with this law of composition $(E_1 \times E_2, \pi_1, \pi_2)$ is a product of the magmas E_1, E_2.

Recall: the universal property of the product set $(E_1 \times E_2, \pi_1, \pi_2)$ is the following: for every pair of mappings

$$p_1 : P \longrightarrow E_1,$$

$$p_2 : P \longrightarrow E_2,$$

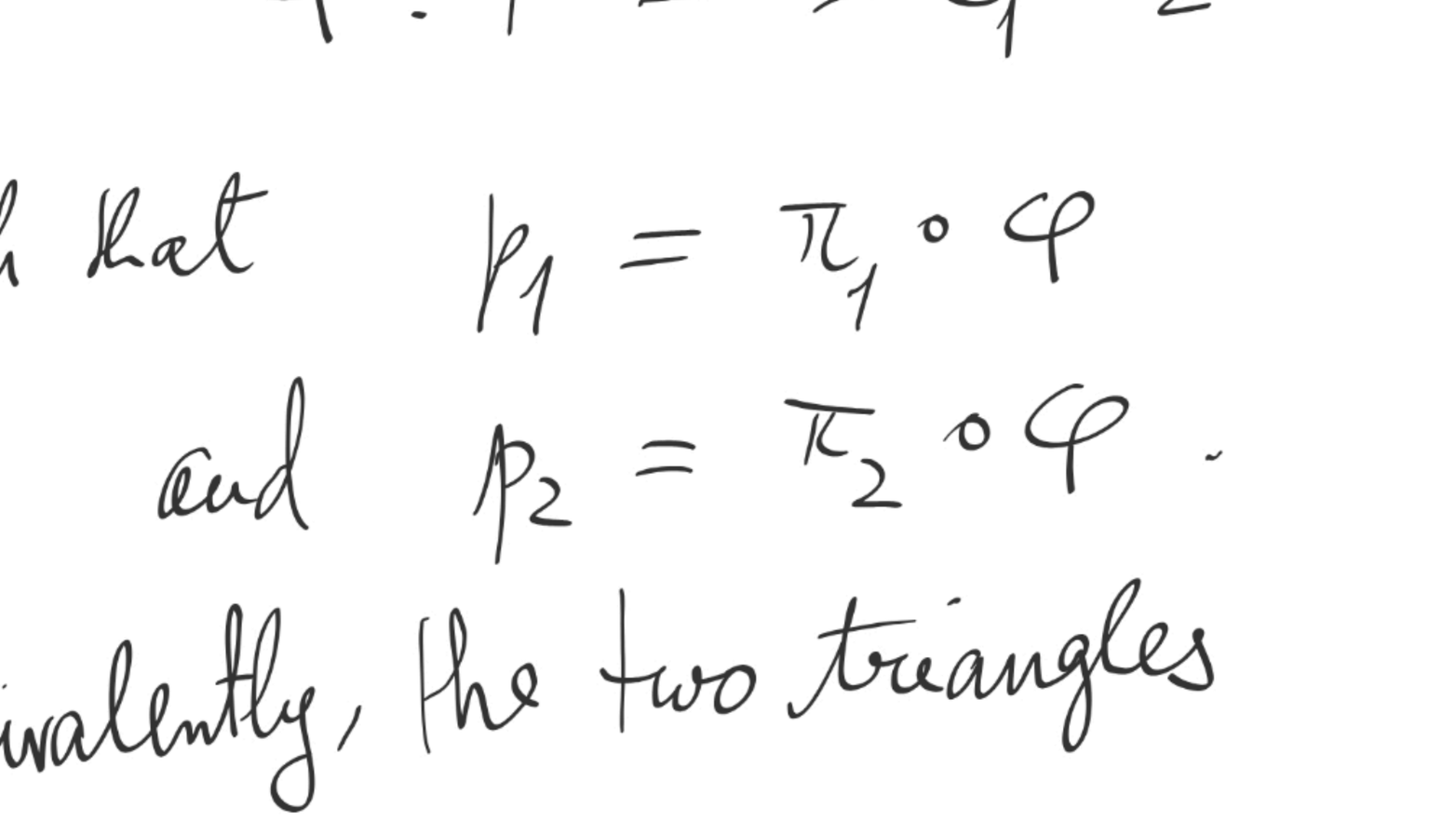
there is a unique mapping

$$\varphi : P \longrightarrow E_1 \times E_2$$

such that $p_1 = \pi_1 \circ \varphi$

and $p_2 = \pi_2 \circ \varphi$.

Equivalently, the two triangles



commute. It is tempting to write $\varphi = (p_1, p_2)$.

Also recall that given two mappings

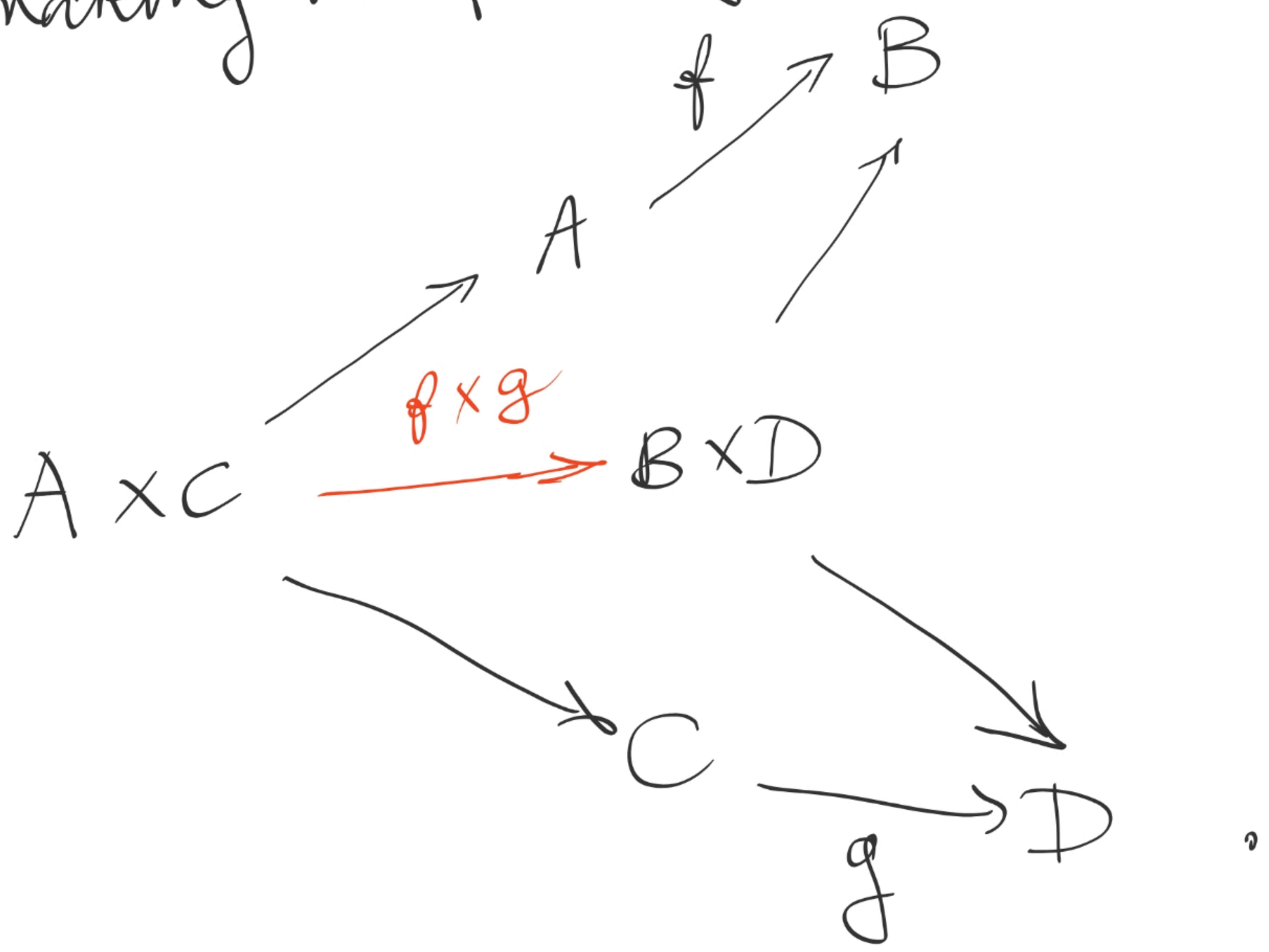
$$f : A \longrightarrow B$$

and $$g : C \longrightarrow D$$

there is a unique mapping

$$f \times g : A \times C \longrightarrow B \times D$$

making the following diagram commute:



Now suppose E_1, E_2 are magmas.

Definition of the law of composition

Let $\mu : (E_1 \times E_2) \times (E_1 \times E_2) \longrightarrow E_1 \times E_2$

$\mu_1 : E_1 \times E_1 \longrightarrow E_1$ and $\mu_2 : E_2 \times E_2 \longrightarrow E_2$

be the laws of composition in E_1 and E_2 respectively.

Consider the mappings

$$(E_1 \times E_2) \times (E_1 \times E_2) \xrightarrow{\pi_1 \times \pi_1} E_1 \times E_1 \xrightarrow{\mu_1} E_1$$

and

$$(E_1 \times E_2) \times (E_1 \times E_2) \xrightarrow{\pi_2 \times \pi_2} E_2 \times E_2 \xrightarrow{\mu_2} E_2 .$$

By the universal property of the product set, there is a unique mapping

$$\mu : (E_1 \times E_2) \times (E_1 \times E_2) \longrightarrow E_1 \times E_2$$

such that $\mu_1 \circ (\pi_1 \times \pi_1) = \pi_1 \circ \mu$

and $\mu_2 \circ (\pi_2 \times \pi_2) = \pi_2 \circ \mu$.

So $(E_1 \times E_2, \mu)$ is a magma.

- π_1 is a morphism of magmas.

We have to check that

$$\begin{array}{ccc} (E_1 \times E_2) \times (E_1 \times E_2) & \xrightarrow{\mu} & E_1 \times E_2 \\ \downarrow{\scriptstyle \pi_1 \times \pi_1} & & \downarrow{\scriptstyle \pi_1} \\ E_1 \times E_1 & \xrightarrow{\mu_1} & E_1 \end{array}$$

Commutes. Indeed, it commutes by construction.

Similarly, π_2 is a morphism of magmas.

The universal property of the product magma

Let P be a magma and $\phi_1 : P \to E_1$, $\phi_2 : P \to E_2$ morphisms of magmas. We have to show that there is a unique morphism of magmas

$$\varphi : P \longrightarrow E_1 \times E_2$$

such that $\phi_1 = \pi_1 \circ \varphi$

and $\phi_2 = \pi_2 \circ \varphi$.

The existence of φ

By the universal property of the product set, there is a unique mapping

$$\varphi : P \longrightarrow E_1 \times E_2$$

such that $p_1 = \pi_1 \circ \varphi$

and $p_2 = \pi_2 \circ \varphi$

We claim that φ is a morphism of magmas.

Equivalently, the square

$$\begin{array}{ccc} P \times P & \xrightarrow{\nu} & P \\ {\scriptstyle \varphi \times \varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ (E_1 \times E_2) \times (E_1 \times E_2) & \xrightarrow{\mu} & E_1 \times E_2 \end{array}$$

commutes, where ν is the law of composition in P. To check the equality

$$\mu \circ (\varphi \times \varphi) = \varphi \circ \nu,$$

it is enough, by the universal property of the product set $E_1 \times E_2$, to check that

$$\pi_1 \circ \mu \circ (\varphi \times \varphi) = \pi_1 \circ \varphi \circ \nu$$

and $$\pi_2 \circ \mu \circ (\varphi \times \varphi) = \pi_2 \circ \varphi \circ \nu.$$

Now, $\pi_1 \circ \varphi \circ \nu = p_1 \circ \nu$,

and, since p_1 is a morphism of magmas,

$$p_1 \circ \nu = \mu_1 \circ (p_1 \times p_1)$$

Also, $\pi_1 \circ \mu \circ (\varphi \times \varphi) = \mu_1 \circ (\pi_1 \times \pi_1) \circ (\varphi \circ \varphi)$

$$= \mu_1 \circ \big((\pi_1 \circ \varphi) \times (\pi_1 \circ \varphi)\big)$$

$$= \mu_1 \circ (p_1 \times p_1).$$

It follows that $\pi_1 \circ \mu \circ (\varphi \times \varphi) = \pi_1 \circ \varphi \circ \nu$

and similarly, $\pi_2 \circ \mu \circ (\varphi \times \varphi) = \pi_2 \circ \varphi \circ \nu$.

Therefore $\mu \circ (\varphi \times \varphi) = \varphi \circ \nu$,

showing that φ is a morphism of magmas.

The uniqueness of φ

This is clear since φ is the unique mapping (let alone morphism of magmas) such that $p_1 = \pi_1 \circ \varphi$ and $p_2 = \pi_2 \circ \varphi$.

This proves that the magma $(E_1 \times E_2, \mu)$

this proves

(equipped with the morphisms π_1, π_2)

is the product of the magmas

(E_1, μ_1) and (E_2, μ_2).